Eclipse Soundscapes Data Management: Receipt, Sorting, & Metadata Organization (Stage 1)

A methods and provenance reference for people using Eclipse Soundscapes (ES) datasets

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1. What This Document Covers

This document describes Stage 1 (Receipt, Sorting, and Metadata Organization) of the Eclipse Soundscapes (ES) data lifecycle. It explains how volunteer-submitted microSD cards, recording-location information, and participant-provided notes were received, organized, and associated with Eclipse Soundscapes identification numbers (ES ID #s) to create structured site-level records for downstream processing.

Eclipse Soundscapes is a NASA-funded participatory science project that deployed AudioMoth acoustic recorders during two eclipses: the October 14, 2023 annular solar eclipse and the April 8, 2024 total solar eclipse. Volunteers recorded environmental soundscapes across the United States so that researchers could study how rapid eclipse-related light changes affected animal behavior and ambient sound.

Stage 1 transformed returned physical media and participant-submitted metadata into organized site-level records that could move forward into later validation, processing, public archiving, and scientific analysis workflows. This stage established the foundational metadata relationships used throughout the remainder of the ES data lifecycle, including the association of recording devices, geographic coordinates, eclipse timing information, and participant documentation with each ES ID # site record.

This document is written as a methods and provenance reference for readers who were not directly involved in the project but need to understand how Eclipse Soundscapes transformed distributed volunteer submissions into organized and traceable datasets. It documents the intake workflows, completeness-based sorting procedures, metadata reconciliation steps, and organizational decisions that prepared records for Stage 2 processing while preserving traceability across the broader ES data stewardship pipeline.

2. What Stage 1 Produced

Stage 1 transformed returned participant materials and submitted metadata into organized, traceable site-level records suitable for downstream processing.

- **Expanded Master Data Collectors Spreadsheet**
  A structured spreadsheet serving as the central metadata record, expanded during Stage 1 to incorporate validated location data, derived eclipse timing and coverage, completeness-based sorting status, and participant-submitted metadata associated with each ES ID #.
- **Organized Site-Level Physical Audio Records**
  Returned microSD cards, envelopes, and handwritten notes organized into ES ID #–based site-level groupings to support traceability, metadata reconciliation, and downstream processing workflows.
- **Analysis Metadata Spreadsheet**
  A streamlined metadata spreadsheet organized by ES ID # and prepared for downstream analysis workflows, containing geographic coordinates and derived eclipse timing information associated with usable recording sites.
- **Eclipse Context Metadata and Timing Derivation (EPTT)**
  Development and operational use of the Eclipse Phase Timing Tool (EPTT), which associated site-level geographic coordinates with NASA eclipse prediction data to derive standardized eclipse timing and eclipse coverage metadata for each ES ID # recording site.

Records lacking sufficient metadata to establish valid site-level records were flagged as incomplete or unusable and removed from downstream processing. The remaining organized records and metadata were prepared for transfer to Stage 2 processing workflows.

3. Why Stage 1 Was Needed

Stage 1 was necessary because Eclipse Soundscapes audio data collection relied on both online metadata submissions and the physical return of audio data via participant-submitted microSD cards. These materials could not be used directly and required a structured intake and reconciliation process to organize returned materials, establish site-level records, associate submissions with the correct ES ID #s, link location information to corresponding audio recordings, and derive eclipse timing and coverage metadata from submitted recording locations.

In practice, submissions varied widely in completeness, format, and packaging. Participants returned materials over an extended period and included a mix of online entries, handwritten notes, printed materials, and inconsistently labeled envelopes. These differences required careful manual review and reconciliation to determine which records contained sufficient information to support downstream processing.

As a result, Stage 1 was not a single intake step but an extended process of organizing, standardizing, sorting, and evaluating submissions to determine which records were complete, which required manual intervention, and which could not be reliably associated with a usable site-level recording record.

Stage 1 established the organized metadata and physical record structure required for downstream validation, processing, public archiving, and scientific analysis workflows. The workflow diagram below provides a Stage 1–focused view of the Eclipse Soundscapes data lifecycle, showing only the directly relevant deliverables from related stages. The complete ES Operational Infrastructure and Data Stewardship Workflows diagram is included in Appendix 1.

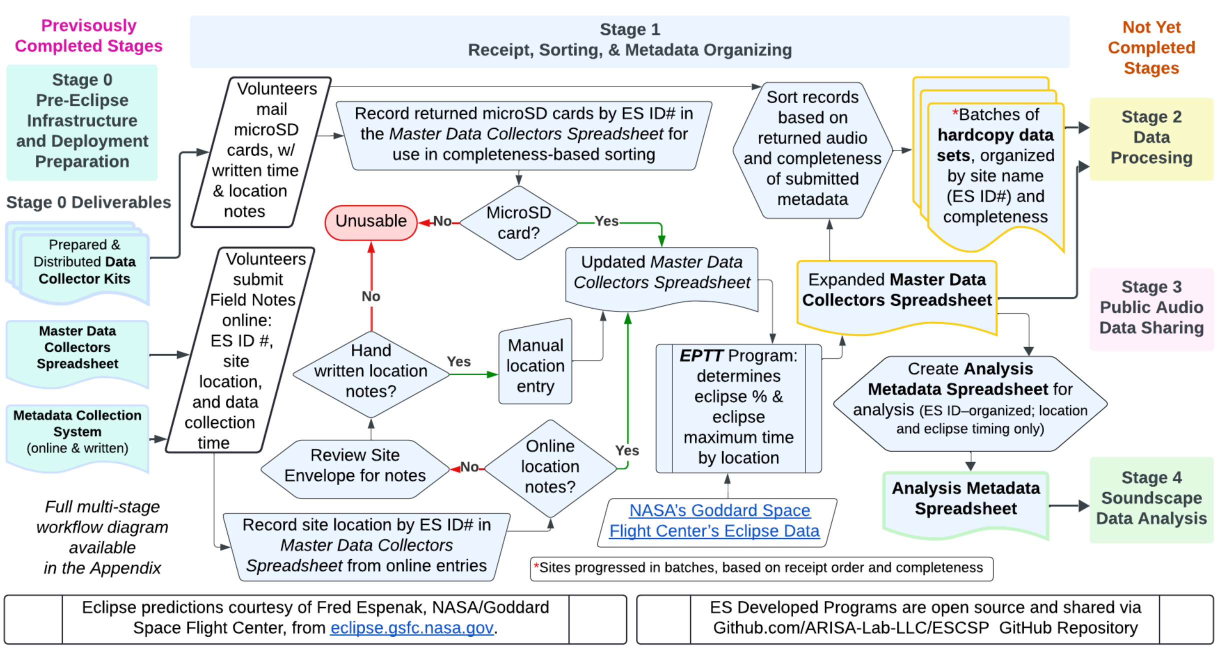

Figure 1. *Eclipse Soundscapes Stage 1 workflow for receipt, sorting, and metadata organization of participant-submitted audio datasets and site information. The workflow shows how physical media, online metadata, handwritten notes, and NASA eclipse timing information derived through the Eclipse Phase Timing Tool (EPTT) were reconciled into organized ES ID# site-level records for downstream processing, public archiving, and scientific analysis.*

4. Data Submission

Stage 1 began with participant submissions, both online and via mail. The project used a submission model designed to be accessible, privacy-aware, and compatible with both project-supplied and participant-owned AudioMoth devices. The basic expectation was that volunteers would return the microSD card containing the audio data recordings and provide the location and timing information needed to interpret those recordings. This volunteer supplied site information was submitted via an online form, and also written notes accompanying the MicroSD card.

The Data collection and submission training and instructions provided to participants are available in this report:

> Severino, M., Winter, H., & Bauer, D. J. (2026). *Eclipse Soundscapes Data Collector Role Training and Implementation Manual (2023–2024)*. Zenodo. https://doi.org/10.5281/zenodo.18623443

4.1 Return of the MicroSD Card

Participants using project-provided kits received returned microSD cards and hand-written field notes using prepaid mailers provided by the project. The AudioMoth recorder itself did not need to be returned; participants could keep the device if they wished or return the device to the ES team to be donated later.

This mailing-based workflow reduced participation barriers. It allowed volunteers to contribute without requiring a computer, a microSD card reader, or reliable high-speed internet connectivity. Returned materials were received and processed as part of stage 1 intake.

4.2 Submission of Recording Location Information

Participants also submitted site information through an online form by entering their ES ID # along with the recording location (latitude and longitude), recording start and stop date and time, and any additional notes that would help interpret the site. This information provided the core metadata needed to connect a returned microSD card to a specific data collection site.

Location data were submitted in decimal degrees, which supported consistency across records and standardized processing. Submitted location information was used in Stage 1 to associate audio recordings with a specific site and to derive eclipse timing and coverage information.

The ES ID # system, described in the Stage 0 documentation, was central to this process, linking submitted site information to returned microSD cards while maintaining separation between participant identity and the scientific data record. The *Master Data Collectors Spreadsheet,* also described in Stage 0 documentation, already contained ES ID # and AudioMoth serial number pairings, and AudioMoth devices automatically embedded their serial numbers within WAV files. Participant-submitted location information, whether from online forms or handwritten notes, was consolidated and verified within the spreadsheet. Programmatic workflows were then used to derive eclipse timing and coverage from site location, followed by ES ID #–based matching of metadata to corresponding audio recordings.

4.3 Differences for Sounds of Nature Participants

Sounds of Nature was an external partner program whose participants also collected eclipse audio data using AudioMoth devices in 2024, when the project expanded to support participation by people who purchased or owned their own AudioMoth devices. Participants in the *Sounds of Nature* partner program followed the same data collection and metadata submission process as other Eclipse Soundscapes participants, including submission of location information through the ES online form using ES ID #s acquired through an ES project developed AudioMoth registration system, described in Stage 0 documentation. The primary difference was in the return of audio data: microSD cards were mailed to the Sounds of Nature coordinator, and the audio recordings were later transferred to the ES team via a hard drive.

Once received, these data were incorporated into the same Stage 1 intake, sorting, and metadata organization workflow as all other submissions. In the final organized record, they are indistinguishable from other submissions except for their earlier audio file submission pathway.

5. Data Intake and Physical Receipt

Stage 1 included the physical intake of returned microSD cards and accompanying materials. Submissions were received over an extended period and varied widely in format and content. In addition to microSD cards, participants often included handwritten notes, printed materials, photographs, and other supplemental information within return envelopes.

Packaging and handling methods also varied considerably. Some submissions included extensively protected materials, such as multiple layers of bubble wrap and tape, which required additional time to unpack and review. Each submission required manual handling to safely access the microSD card and associated materials while preserving all included information.

As part of intake, each returned package was opened and reviewed using the ES ID # written on the package and accompanying materials, as participants had been instructed to label both the return envelope and any enclosed notes with their assigned ES ID #. The corresponding record in the *Master Data Collectors Spreadsheet* was then reviewed.

If a microSD card was present, the spreadsheet was updated to indicate that the microSD card had been physically received. At that time, the spreadsheet was also checked to determine whether location metadata (latitude and longitude) had already been received through the participant's online form submission process, which had been incorporated into the spreadsheet before physical sorting of returned materials.

If sufficient location metadata was already present in the spreadsheet, the returned materials were separated into two organized physical groupings. Supplemental materials not required for Stage 2 processing, such as photographs, printed materials, and other participant-provided items, were retained in a separate archival filing system associated with the ES ID #. The returned microSD card and any location or timing notes relevant to downstream processing were placed into a standardized envelope system organized by ES ID #. This standardized packaging approach was used to simplify and streamline the later manual microSD card ingestion and upload workflow performed during Stage 2 processing.

Standardizing the physical organization of microSD cards and associated notes reduced handling complexity during Stage 2 and supported more efficient batch-based processing of returned audio data.

If location metadata was not present in the spreadsheet, indicating that an online form submission had not been received or matched to that ES ID #, handwritten notes and supplemental materials included in the returned package were reviewed for usable location information. When latitude and longitude information were available in the handwritten materials, it was manually entered into the *Master Data Collectors Spreadsheet*.

If a returned microSD card still lacked sufficient location metadata after review of the physical materials, the record was placed into a separate manual investigation group. These records underwent additional reconciliation efforts, including review of participant contact information, follow-up outreach when possible, and additional project-wide reminders requesting submission of missing online location information. Records that ultimately could not be associated with sufficient site-level metadata remained flagged as incomplete or unusable within the Stage 1 workflow.

Following intake and reconciliation, materials were reviewed to confirm whether a microSD card was physically present and whether sufficient location information was available to establish a site-level record for Stage 2. The contents of the microSD card were not inspected during Stage 1; audio file presence, readability, and device metadata verification occurred later during Stage 2 processing. Variability in packaging, labeling, and included materials required careful handling and organization to ensure that all relevant information was captured and associated correctly.

This intake process was time-intensive and formed a critical first step in organizing and standardizing participant submissions prior to sorting and metadata organization.

6. Data Sorting

Following data intake and initial consolidation of submitted materials, records were evaluated and sorted based on the completeness of returned audio and submitted metadata. The purpose of sorting was to determine which submissions contained sufficient information to establish a valid site-level record and which required additional review or could not be used. Because participant submissions varied widely in format, labeling, and completeness, this step was essential for organizing records into structured groups prior to further processing.

6.1 Sorting Criteria

Sorting decisions were based on the presence of core elements required to establish a site-level record. A usable record required:

- A returned microSD card.
- A valid ES ID # is associated with the submission.
- Sufficient location information to associate the data recording with a specific data collection site.

Location information was typically obtained through participant-submitted online field notes and, when necessary, supplemented by handwritten notes enclosed with returned materials. These elements collectively enabled the creation of a coherent and traceable record for each data collection site. Records lacking sufficient location metadata, even when a microSD card was returned, were not considered usable site-level records and were marked as “unusable.”

6.2 Recording MicroSD Card Returns

As part of the sorting process, the *Master Data Collectors Spreadsheet* was updated to record the presence of returned microSD cards for each ES ID #. The absence of a recorded return was interpreted as the microSD card not being received. This approach provided a consistent and systematic method for identifying records with available audio data and supported completeness-based sorting across all submissions.

6.3 Sorting Categories

Based on the criteria above, records were grouped into categories reflecting their completeness and readiness for further processing:

- **Usable**: Records containing a returned microSD card and sufficient location information to establish a site-level record.
- **Requires Manual Review**: Records with missing, inconsistent, or unclear location metadata, requiring reconciliation using online submissions and handwritten notes. Manual review in Stage 1 focused specifically on resolving metadata needed to establish a valid site-level record.
- **Unusable**: Records lacking the minimum information required to associate audio data with a specific site, including cases where a microSD card was not returned, and no reliable metadata was available.

Sorting decisions were based on the information available at the time of intake and did not attempt to fully resolve all inconsistencies. Instead, this step organized records into structured groups that could be systematically addressed in subsequent processing steps. Manual review and reconciliation efforts in Stage 1 were limited to location information and did not involve inspection or processing of the audio data.

6.4 Batch Progression Through the Pipeline

Sites progressed through the Stage 1 pipeline in batches based on receipt order and completeness. This batching approach allowed more complete records to be prioritized for processing, while incomplete or ambiguous submissions were set aside for later review. Managing records in this way supported an efficient workflow while maintaining data integrity and traceability. These sorted groupings, organized by site (ES ID #) and completeness status, formed the basis for downstream handling and were passed to Stage 2 for audio ingestion, validation, timestamp review, and secure backup. Records that included a returned microSD card but lacked sufficient location metadata were separated from fully usable site-level records, as they could not yet be reliably interpreted or linked to a specific data collection site.

7. Metadata Organization

Following data sorting, metadata were organized to create standardized, site-level records that could support downstream processing. This step integrated information from multiple sources into the *Master Data Collectors Spreadsheet*, which was established in Stage 0 and served as the central metadata management system throughout the Eclipse Soundscapes data lifecycle.

7.1 Consolidation of Metadata Sources

Following data sorting, metadata from multiple sources were consolidated to create standardized, site-level records associated with each ES ID #. These sources included participant online submissions, handwritten notes, returned physical materials, pre-existing AudioMoth serial number assignments established during Stage 0, and eclipse-context data derived during Stage 1.

The Master Data Collectors Spreadsheet served as the central and evolving system of record throughout this process, linking physical audio media, participant-submitted metadata, and derived eclipse-context information into standardized site-level records that could support downstream validation, analysis, public sharing, and long-term stewardship.

7.2 Role of the Preliminary *Master Data Collectors Spreadsheet*

The Preliminary *Master Data Collectors Spreadsheet* was established during Stage 0, when AudioMoth device serial numbers were assigned to ES ID numbers before device distribution. This assignment occurred both for project-provided devices and through a registration process for participants who used their own AudioMoth devices. As a result, the spreadsheet initially served as a registry linking each ES ID # to a specific AudioMoth device.

During Stage 1, the spreadsheet was expanded to incorporate metadata derived from participant submissions and intake processes. For each ES ID #, the spreadsheet included:

- Confirmation of the presence of the returned microSD card.
- Associated AudioMoth serial number.
- Geographic coordinates of the recording site.
- Recording start date and time, when available.
- Participant-provided site notes.
- Indicators of completeness or the need for manual review.
- Derived eclipse context (added during Stage 1).

By integrating both pre-collection device assignments and post-collection metadata, the spreadsheet ensured consistent linkage between participant submissions and corresponding audio recordings. **The *Master Data Collectors Spreadsheet* functioned as the central and evolving system of record throughout the Eclipse Soundscapes data lifecycle**, supporting subsequent validation, processing, and archiving activities. Where possible, missing or inconsistent location metadata were resolved during Stage 1 through manual reconciliation using online submissions and handwritten notes, ensuring that the *Master Data Collectors Spreadsheet* reflected the most complete and accurate site-level information available.

7.3 Derivation of Eclipse Context

Once location information was consolidated within the *Master Data Collectors Spreadsheet*, the Eclipse Soundscapes team used the Eclipse Phase Timing Tool (EPTT) to derive eclipse-specific context for each data collection site. Using reference data from NASA's Goddard Space Flight Center (https://eclipse.gsfc.nasa.gov/eclipse.html), eclipse timing and coverage data were calculated based on the geographic coordinates associated with each ES ID #.

The Eclipse Phase Timing Tool (EPTT) was developed by the Eclipse Soundscapes team to associate NASA eclipse prediction data with site-level geographic coordinates (further details in the Eclipse Soundscapes GitHub repository - https://github.com/ARISA-Lab-LLC/ESCSP). By incorporating standardized eclipse timing and eclipse coverage information directly into the Master Data Collectors Spreadsheet, the workflow enabled downstream researchers outside the field of eclipse science, including soundscape ecology and acoustic analysis researchers, to work with scientifically aligned eclipse-context metadata associated with each recording site.

The following eclipse parameters were derived and incorporated into the spreadsheet for each site:

- **Eclipse Type**
    - Total
    - Partial
- **Eclipse Percent**
    - The percentage of solar obscuration at the data collection site.

- **Eclipse Start (UTC – First Contact)**
  - The time at which the Moon first begins to obscure the Sun.
- **Totality Start (UTC – Second Contact)** (for sites within the path of totality)
  - The time at which totality begins.
- **Eclipse Maximum (UTC)**
  - The moment of greatest solar obscuration at the site.
- **Totality End (UTC – Third Contact)** (for sites within the path of totality)
  - The time at which totality ends.
- **Eclipse End (UTC – Fourth Contact)**
  - The time at which the Moon completely stops blocking the Sun.

For sites experiencing only a partial eclipse, the totality-related fields (Second and Third Contact times) were left blank or recorded as not applicable. All times were recorded in Coordinated Universal Time (UTC) to ensure consistency across the geographically distributed dataset.

These derived eclipse timing and obscuration data were incorporated into each site-level record, ensuring that every dataset included standardized contextual information. This enabled consistent interpretation of the acoustic recordings and supported downstream validation and analysis. The use of NASA-derived eclipse circumstances ensured scientific accuracy and reproducibility across the entire dataset.

7.4 Finalization of the *Master Data Collectors Spreadsheet*

Following the consolidation of participant-submitted metadata and the derivation of eclipse context, the *Master Data Collectors Spreadsheet* was updated to reflect the completed outcomes of Stage 1. Originally established during Stage 0 to link ES ID numbers with AudioMoth serial numbers, this same spreadsheet was expanded throughout Stage 1 to incorporate all additional metadata associated with each data collection site.

Rather than creating a separate document, Stage 1 resulted in a finalized version of the existing *Master Data Collectors Spreadsheet*. This version represented a standardized and quality-checked record of all site-level metadata at the conclusion of Stage 1 and served as the authoritative reference for subsequent stages of validation, processing, archiving, and public dissemination.

7.4.1 Contents of the *Master Data Collectors Spreadsheet*

For each **ES ID #**, the *Master Data Collectors Spreadsheet* included the following fields:

- **ES ID #**: Unique identifier assigned to each AudioMoth device and data collection site.
- **AudioMoth Serial Number**: Device serial number associated with the ES ID #, established during Stage 0.
- **microSD Card Return Status**: Confirmation of the presence of a returned microSD card (absence inferred if not recorded).
- **Sorting Status**: Classification of each record as Usable, Requires Manual Review, or Unusable. This status was iterative and could evolve as additional information became available through manual review and reconciliation.
- **Latitude and Longitude:** Verified geographic coordinates of the recording site in decimal degrees.
- **Recording Start Date and Time (UTC)**: Included only when provided through the online field notes submission. Timing information present solely in handwritten notes was not transcribed into the spreadsheet during Stage 1.

- **Participant Site Notes**: Additional contextual information provided by participants through online or handwritten submissions.
- **Eclipse Type**: Classification of the eclipse at the site (Total or Partial).
- **Eclipse Percent**: Percentage of solar obscuration at the data collection site.
- **Eclipse Start (UTC - First Contact)**: Time at which the Moon first begins to obscure the Sun.
- **Totality Start (UTC - Second Contact):** Time at which totality begins (applicable to sites within the path of totality).
- **Maximum Eclipse Time (UTC)**: Time of greatest solar obscuration.
- **Totality End (UTC - Third Contact)**: Time at which totality ends (applicable to sites within the path of totality).
- **Eclipse End (UTC - Fourth Contact)**: Time at which the Moon completely leaves the solar disk.

For sites experiencing only a partial eclipse, the totality-related fields were recorded as not applicable.

7.4.2 Role and Use of the Stage 1 *Master Data Collectors Spreadsheet*

The Stage 1 version of the *Master Data Collectors Spreadsheet* served as the authoritative Stage 1 deliverable, providing a standardized and traceable metadata record for each data collection site. Each record incorporated both participant-submitted information and NASA-derived eclipse attributes, ensuring that every dataset included consistent temporal and contextual information.

The *Master Data Collectors Spreadsheet*, established during Stage 0, was expanded and finalized during Stage 1 to incorporate participant-submitted metadata, evolving sorting classifications, and NASA-derived eclipse context, providing a standardized site-level record for each ES ID #. Records that could not be resolved to include sufficient location metadata were retained but remained flagged, and were not considered complete site-level records at the conclusion of Stage 1.

7.4.3 Creation of the *Analysis Metadata Spreadsheet*

After Stage 1 reconciliation and metadata organization were completed, a separate *Analysis Metadata Spreadsheet* was manually prepared from the updated *Master Data Collectors Spreadsheet*. This process occurred after returned packages had been received and sorted, follow-up efforts for missing location information had been completed, and eclipse timing and coverage information had been derived for usable recording sites using the Eclipse Phase Timing Tool (EPTT) workflows.

The *Analysis Metadata Spreadsheet* was organized by ES ID #, matching the naming structure used for the corresponding audio datasets. Each row represented a single usable recording site and included standardized site-level geographic coordinates and derived eclipse timing information associated with that dataset.

This spreadsheet excluded participant-identifying information and broader intake metadata and was specifically structured to support downstream Stage 4 analysis workflows by providing the analysis team with consistent site location and eclipse-context metadata associated with each usable dataset.

7.5 Organization and Handoff of Physical Audio Media

Following sorting and metadata organization, returned microSD cards were organized into structured, site-level groupings based on ES ID #. Each microSD card represented the original storage medium for the raw audio recordings collected by participants.

At this stage, no extraction, inspection, or modification of the audio data was performed. All recordings remained stored on the microSD cards exactly as recorded by the AudioMoth devices, preserving embedded device metadata and original file structures.

The organization of physical audio media was aligned with the *Master Data Collectors Spreadsheet*, ensuring that each microSD card could be reliably associated with its corresponding site-level metadata record through the ES ID #. Supporting materials, including handwritten notes and packaging information, were used as needed to confirm or reconcile these associations.

Manual review conducted in Stage 1 focused on resolving metadata issues, particularly those related to location information. This included reconciling discrepancies between online submissions and handwritten notes and standardizing all location data into decimal degree format where needed. This review was completed before handoff, resulting in a finalized set of site-level records with consistent and usable metadata.

Returned microSD cards that still lacked sufficient location metadata after Stage 1 review were retained but flagged as unusable and removed from further progression through the data pipeline, as they could not be reliably associated with a defined data collection site.

The remaining organized microSD cards, representing usable site-level records, were transferred to Stage 2 for audio ingestion, validation, timestamp review, and secure backup.

Key characteristics of this Stage 1 output included:

- Preservation of original audio data in its native format on microSD cards
- No alteration or processing of audio files
- Organization of physical media by site (ES ID #)
- Traceable linkage between physical audio data and metadata records
- Completion of metadata review and standardization (including decimal degree formatting) prior to handoff
- Removal of unusable records from the processing pipeline

8. Glossary

This glossary is shared across the Eclipse Soundscapes Stage 0–3 data management documents and defines terminology used throughout the ES data lifecycle workflow.

Term	Definition
AME / ES AMES	ES AudioMoth Metadata Extractor Suite. A software tool used to extract and evaluate embedded AudioMoth metadata, including timestamps, serial numbers, firmware information, and recording details to support validation and provenance workflows.
annular solar eclipse	when the Moon blocks the center of the Sun from view. The Moon appears to be too small to completely block the Sun.
API	Application Programming Interface. A structured method that allows software systems to communicate programmatically with external platforms or services.
Arbimon	An acoustic data platform originally considered by Eclipse Soundscapes for data storage, browsing, and analysis prior to the project's transition to Zenodo.
AudioMoth	An open-source, low-cost acoustic recording device used by ES volunteers to capture environmental soundscapes during eclipse events.
AZUS	Automated Zenodo Upload Software. A configuration-driven, open-source software pipeline developed by ARISA Lab to prepare, package, document, validate, and upload Eclipse Soundscapes datasets to Zenodo.
batch processing	A workflow approach that processes large groups of datasets simultaneously rather than handling one dataset at a time.
CONFIG.TXT	An AudioMoth-generated configuration file containing device settings such as sample rate, gain, firmware version, serial number, recording schedule, temperature, and battery state.
CSV	Comma-Separated Values. A simple, human- and machine-readable text file format used for spreadsheets and structured tabular data.
data dictionary	A structured reference file defining variable names, descriptions, data types, units, sources, code meanings, and notes associated with a dataset or metadata file.
dataset DOI / data site DOI	A persistent Digital Object Identifier assigned to a published Zenodo dataset record, allowing the dataset to be cited, accessed, and reused over time.
dry-run mode	A validation mode used by AZUS to verify configuration settings, metadata, credentials, and file paths before live uploads occur.

eclipse contact times	The calculated timing of eclipse phases for a specific site, including first contact, second contact, maximum eclipse, third contact, and fourth contact.
eclipse maximum	The moment of greatest solar obscuration at a specific location during a solar eclipse.
eclipse percent	The percentage of solar obscuration at a specific location.
ES ID #	Eclipse Soundscapes Identification Number. A unique numeric identifier assigned to each recording site, allowing datasets to be publicly shared without exposing participant identity.
ES WAVES	Eclipse Soundscapes WAV Audio Validation & Extraction Suite. A software system that ingests audio data from multiple microSD cards simultaneously, validates WAV files, extracts metadata, organizes datasets, and supports downstream analysis workflows.
FAIR principles	A framework for scientific data stewardship emphasizing that datasets should be Findable, Accessible, Interoperable, and Reusable.
file_list.csv	A manifest file included in each Zenodo record that lists every file, file size, file type, description, SHA-512 hash, and associated data dictionary.
first contact	solar eclipse Start (First Contact). The time at which the Moon first begins to obscure the Sun.
fourth contact	Eclipse End (Fourth Contact) The time at which the Moon completely stops blocking the Sun from view.
Hash / SHA-512 Hash	A cryptographic fingerprint used to verify file integrity. ES used SHA-512 hashes to confirm that files were not altered or corrupted during storage, transfer, or download.
Human-In-The-Loop review	A workflow step requiring manual human review and interpretation when automated validation systems identify incomplete, inconsistent, or ambiguous data conditions.
HTML README template	A reusable template used to generate standardized, site-specific Zenodo record descriptions and README documentation through automated metadata substitution.
InvenioRDM	The open-source digital repository framework underlying Zenodo. AZUS communicates with Zenodo through the InvenioRDM REST API.

JBOD	"Just a Bunch of Disks." A direct-attached storage system consisting of multiple hard drives used for scalable local data storage.
JSON configuration file	A human-readable configuration file used by AZUS to define project metadata, upload settings, templates, required metadata fields, file paths, and workflow behavior.
Level 0 data	Raw, unprocessed data preserved in its original form without timestamp correction, resampling, filtering, or modification.
metadata	Structured descriptive information about data, including location, timestamps, eclipse parameters, device information, and processing documentation.
metadata reconciliation	The process of comparing and integrating metadata from multiple sources to resolve inconsistencies and establish validated site-level records.
microSD Card	A removable flash memory storage device used within AudioMoth recorders to store WAV audio recordings and configuration files.
participatory science	Scientific research conducted with participation from members of the public contributing observations, data collection, or analysis activities.
PII	Personally Identifiable Information. Protected personal information that could uniquely identify an individual participant.
provenance	Documentation describing the origin, processing history, validation status, and stewardship history of a dataset.
README	Human-readable documentation accompanying a dataset that explains its contents, structure, collection methods, metadata, and usage guidelines.
related identifiers	Structured citation metadata linking Zenodo records to related datasets, publications, software repositories, reports, or videos.
REST API	A web-based application programming interface that allows software systems to communicate using standardized HTTP requests.
scalability	The ability of a workflow, infrastructure system, or software tool to efficiently support increasing amounts of data, users, or processing demand.
second contact	Totality Start (for sites within the path of totality). The time at which totality begins during a total solar eclipse.
site-level dataset	A complete dataset associated with a single Eclipse Soundscapes recording location identified by one ES ID #.

staging directory	A temporary preparation directory created during AZUS processing that contains the ZIP archive and all companion files ready for upload to Zenodo.
subject labels	Standardized Zenodo metadata tags used to categorize datasets by eclipse event, analysis relevance, site type, or project designation.
template substitution	An automated process in which variable placeholders within templates are replaced with site-specific metadata values during dataset generation.

9. References and Related Resources

These references and related resources are shared across the Eclipse Soundscapes (ES) Stage 0–3 data management documents and provide supporting literature, technical resources, software references, infrastructure documentation, and related materials referenced throughout the ES data lifecycle workflow.

Core Eclipse Soundscapes Resources

Eclipse Soundscapes Zenodo Community
Public archive of Eclipse Soundscapes datasets, documentation, and related resources.
https://zenodo.org/communities/eclipsesoundscapes

ARISA Lab Eclipse Soundscapes GitHub Repository
Open-source software, workflows, processing tools, and supporting documentation for the Eclipse Soundscapes project. https://github.com/ARISA-Lab-LLC/ESCSP

Eclipse Soundscapes Technical Workflow References

Severino, M., & Winter, H. (2026). Eclipse Soundscapes Data Collector Role Training and Implementation Manual (2023–2024). Zenodo. https://doi.org/10.5281/zenodo.18623443

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Eclipse Timing and Astronomical Resources

NASA Goddard Space Flight Center Eclipse Web Site
Eclipse timing predictions and eclipse path calculations courtesy of Fred Espenak, NASA/GSFC.
https://eclipse.gsfc.nasa.gov/eclipse.html

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Open Science and Repository Infrastructure Resources

Zenodo
General-purpose open-access repository operated by CERN and OpenAIRE for long-term data preservation and DOI assignment. https://zenodo.org

FAIR Principles
Findable, Accessible, Interoperable, and Reusable data stewardship principles. https://www.go-fair.org/fair-principles/

InvenioRDM
Open-source repository framework underlying Zenodo and used by AZUS through the REST API. https://inveniordm.docs.cern.ch/

Zenodo REST API Documentation
Documentation for Zenodo's Open API used by AZUS for automated dataset publication workflows.
https://developers.zenodo.org/

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Appendix 1: ES Operational Infrastructure and Data Stewardship Workflows